

Hands-on Learning Activity (Lesson 2)

This activity is designed to test and reinforce your knowledge gained in the Lesson 2.

This activity is optional and based on what you learned in the previous two videos you are requested to try-out these steps and share your results in the format with others for peer review.

Overview

This hands-on activity is designed to deepen your understanding of applying Generative AI (GenAI) in algorithmic trading. By integrating GenAI tools, you will explore strategies such as Supertrend-based market direction analysis, gain insights into historical performance, and evaluate practical trading strategies. The activity encourages peer-to-peer collaboration and builds your ability to analyze trends and make data-driven decisions in financial markets.

This assignment aligns with the broader goal of using Generative AI to optimize trading strategies, portfolio management, and financial research.

Learning Objectives

By completing this assignment, students will:

Apply GenAI Models for Market and Financial Analysis:

Analyze historical market data and generate actionable insights for trading decisions using GenAI.

Design GenAI-Based Trading Strategies to Test Trading Hypotheses:

Create, implement, and evaluate trading strategies using tools like Supertrend, incorporating GenAI techniques.

Project Steps

1. Set up your python environment and install all required libraries.

2. Explore and Analyze Market Direction with Supertrend. Try using the GenAI and chatGPT to get your codes:

→ 1. How the Supertrend indicator works in identifying market trends?

→ 2. Get historical data using yahoo finance and get values of supertrend

→ 3. Make a simple strategy to buy the stock when supertrend is green and sell when supertrend is red.



4. Analyse the historical performance of supertrend.

5. Please summarize any significant findings of the strategy.

3. Analyse and give your recommendations about the Supertrend strategy to the community.

Please download your output in the PDF or XLSX and upload in the peer-review section for review. You are also requested to review the output of fellow learners uploaded here.